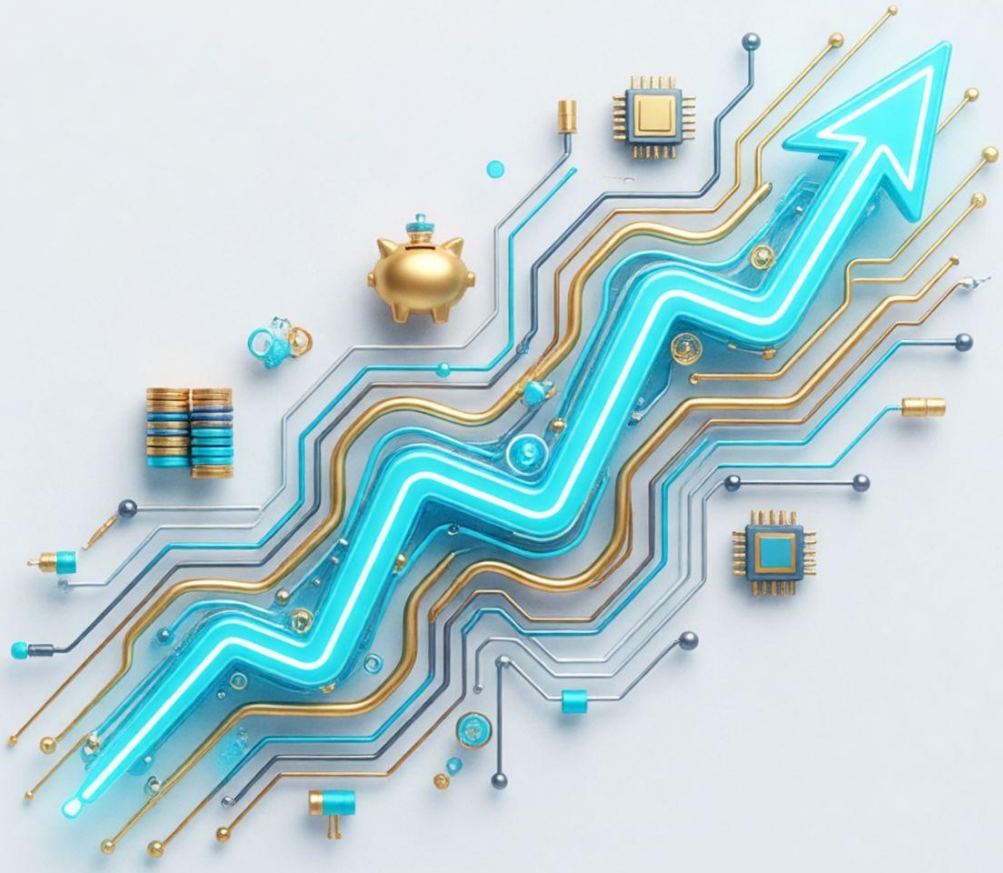
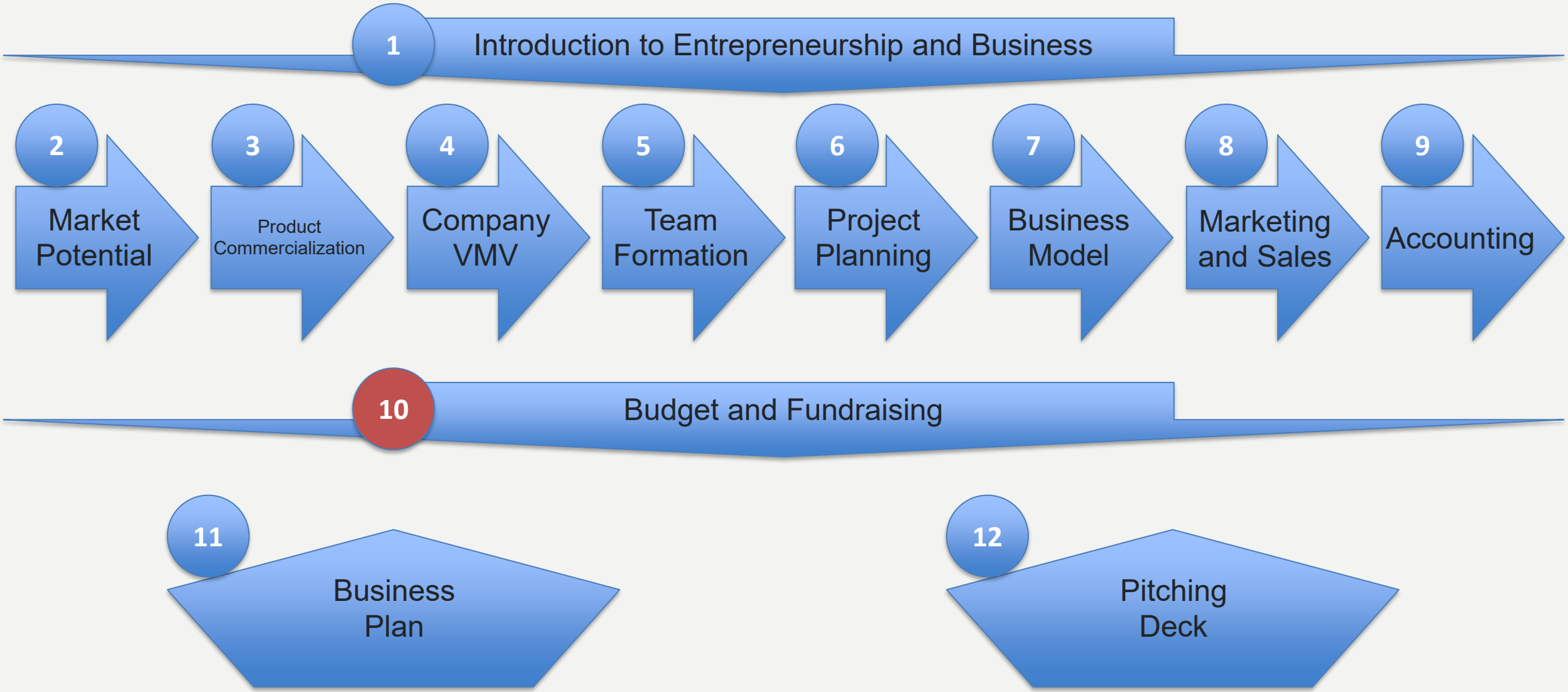
Financial Planning and Capital Acquisition Strategies for Tech Ventures in Electrical and Electronic Engineering

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Outline of the Course



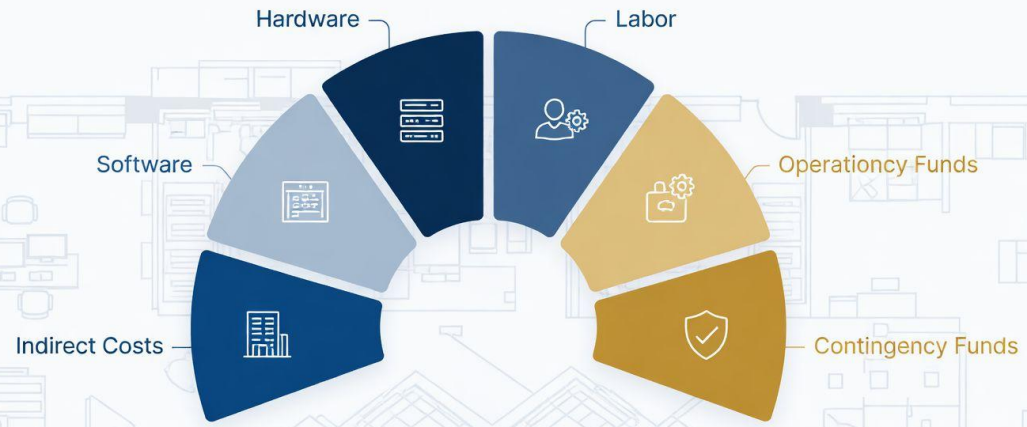
Budget Plan for Technology-based Project

Ir Prof Alan Lam, JP

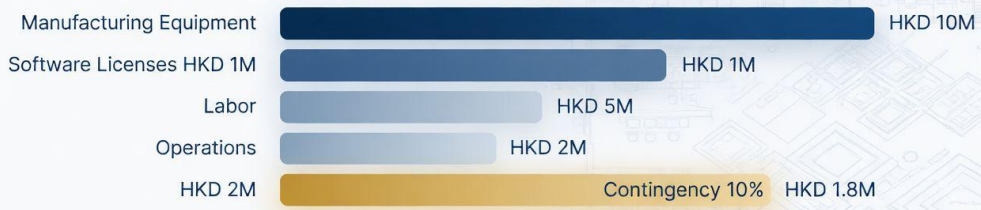


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Understanding Budget Components



Practical Budgeting Example



Cost Behavior



Market Dynamics

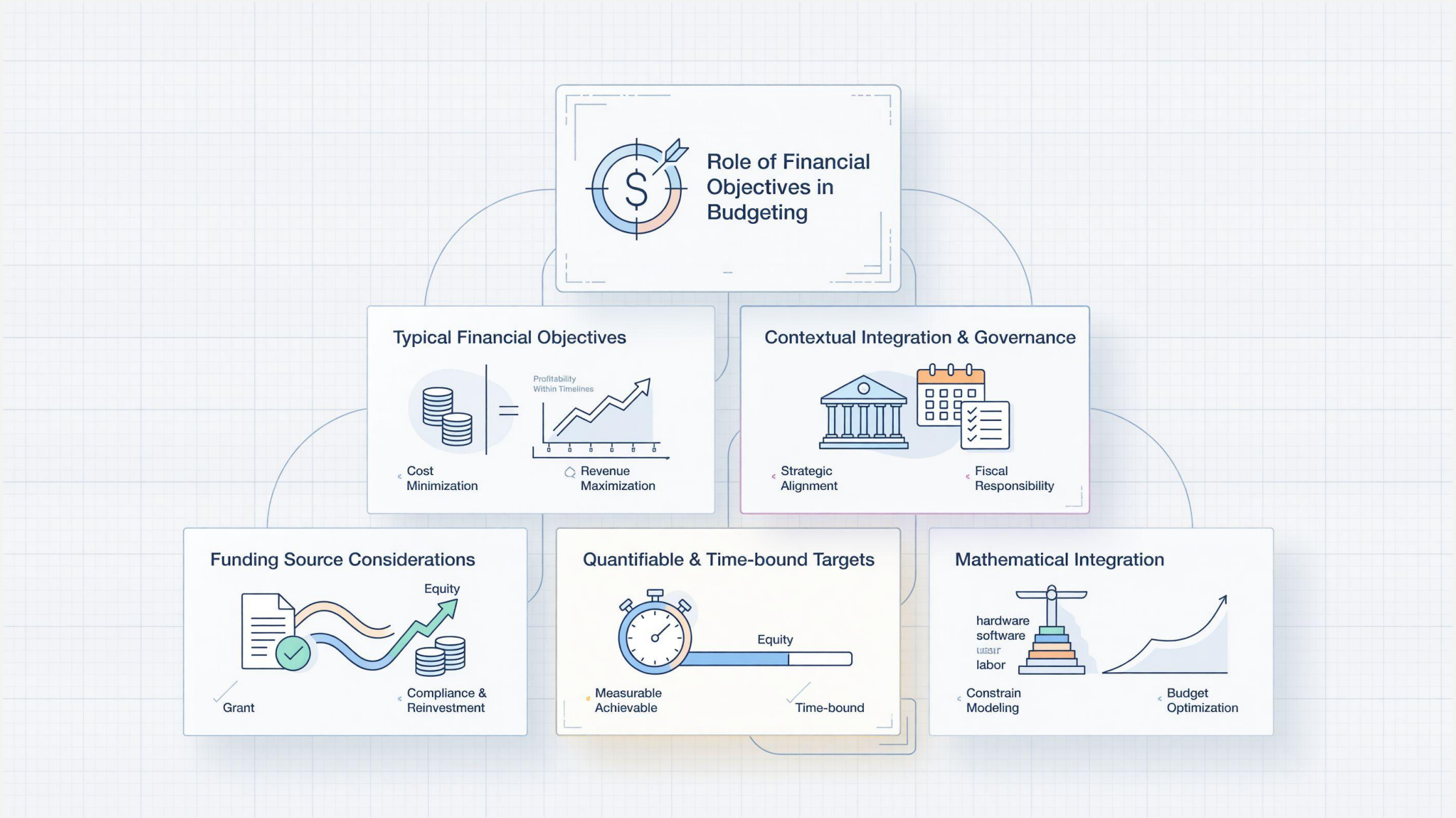


Budget Management Review

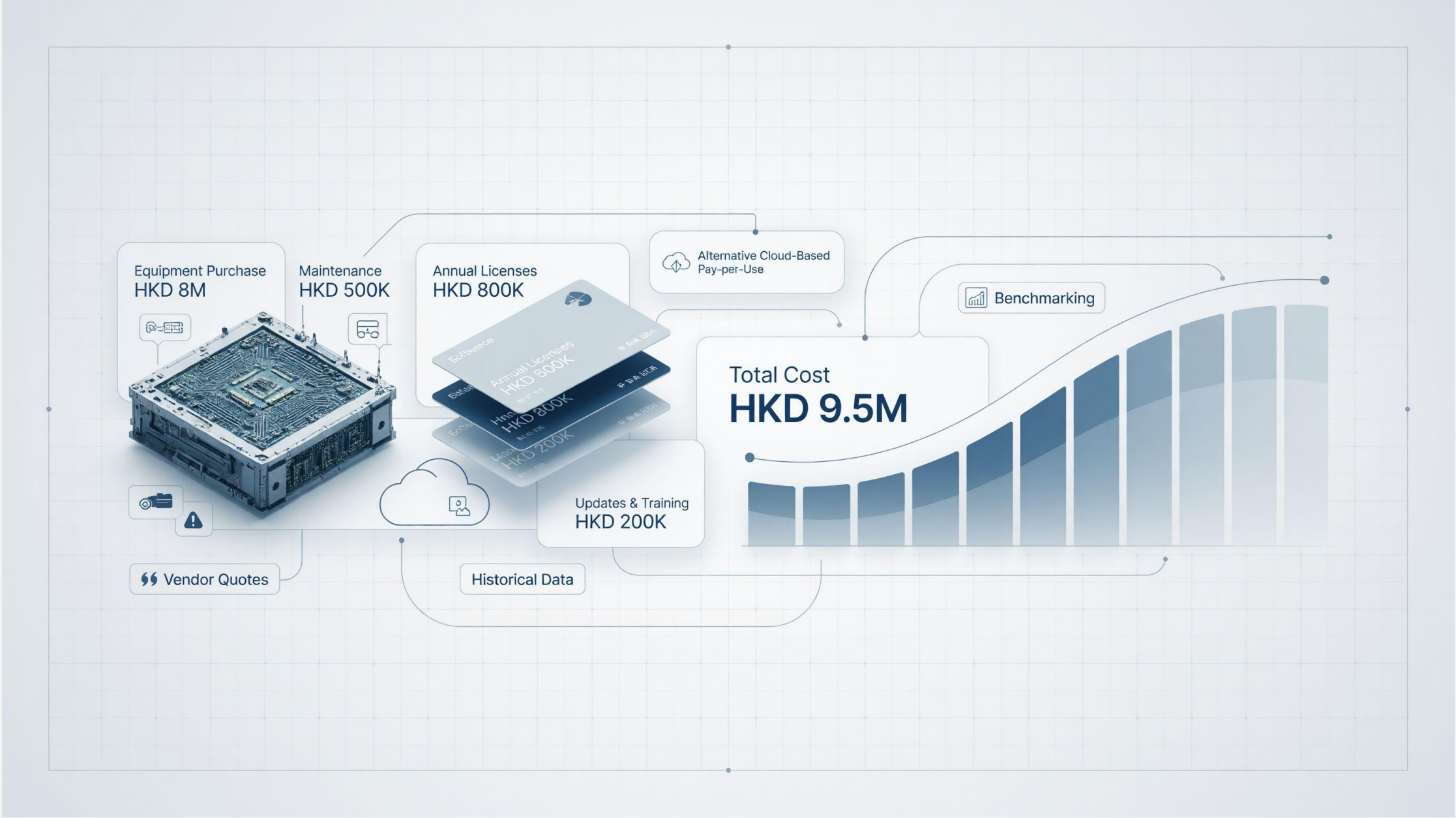
Refine Allocations

Control Deviations

Setting Financial Objectives



Estimating Hardware and Software Costs



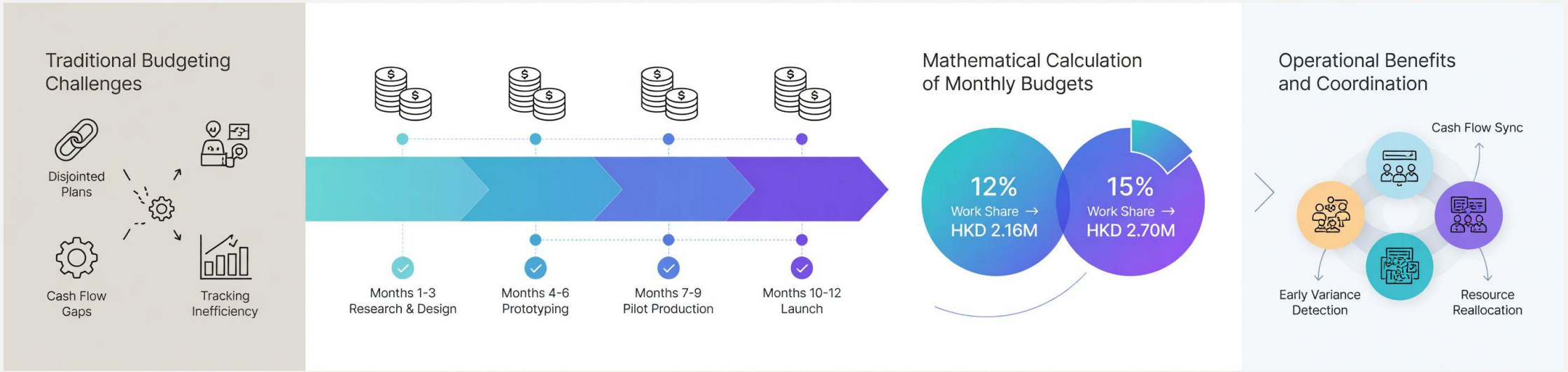
Calculating Labor and Operational Expenses



Allocating Contingency Funds



Creating a Project Timeline Budget



Reviewing Market Benchmarks for Budgeting

Importance of Market Benchmarking

- Competitiveness
- Feasibility
- Accuracy



Benchmarking Sources and Examples



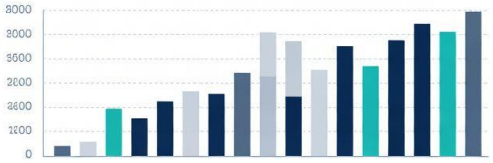
Procurement, Industry, Government.

Benchmarking Methods Across Cost Categories



Practical Salary Benchmarking Example





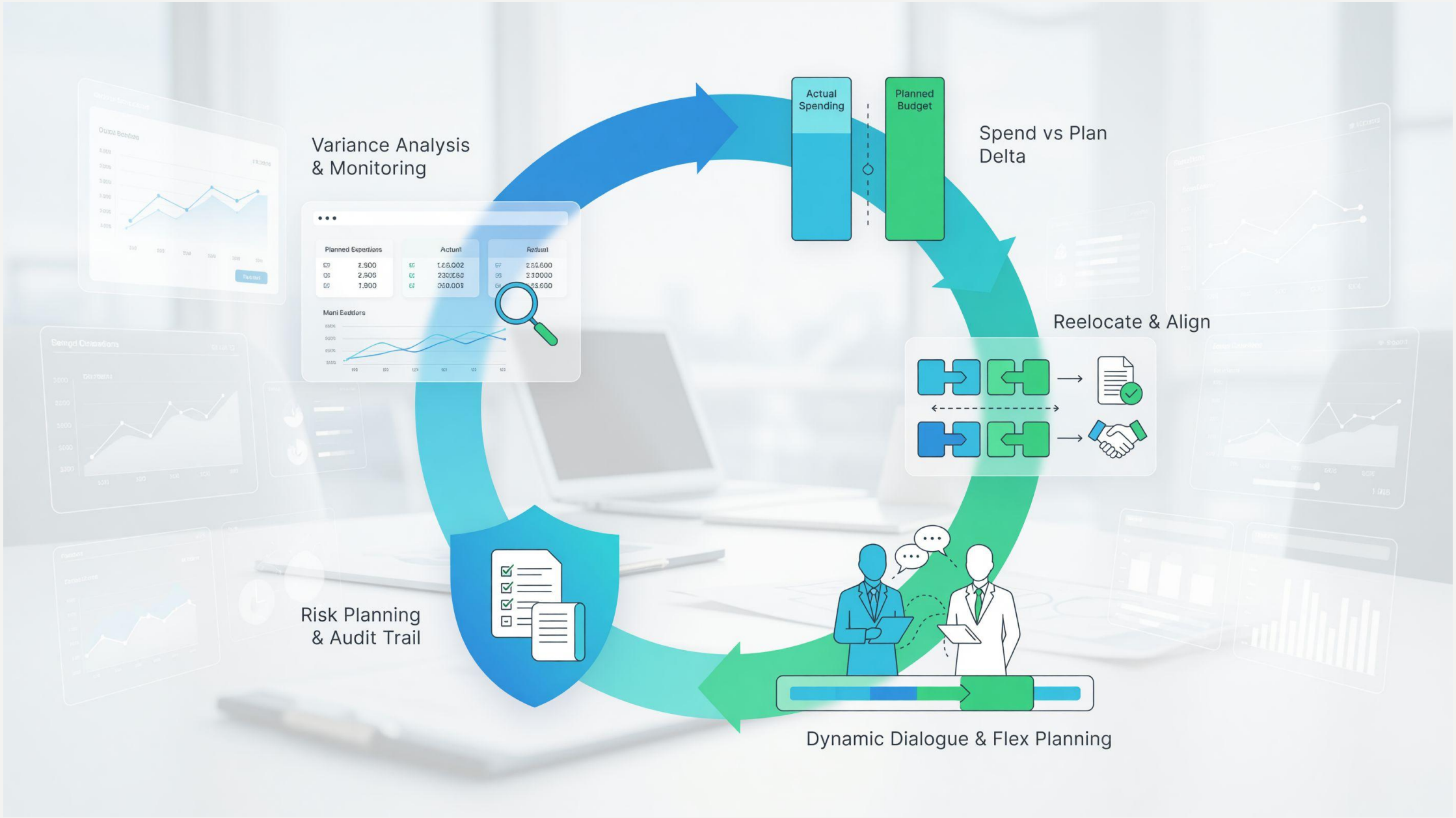
HK Junior Engineers: 28K–32K vs 40K Budget

Data Quality & Ongoing Relevance



Authoritative • Updated • Adaptive

Budget Tracking and Adjustment Techniques



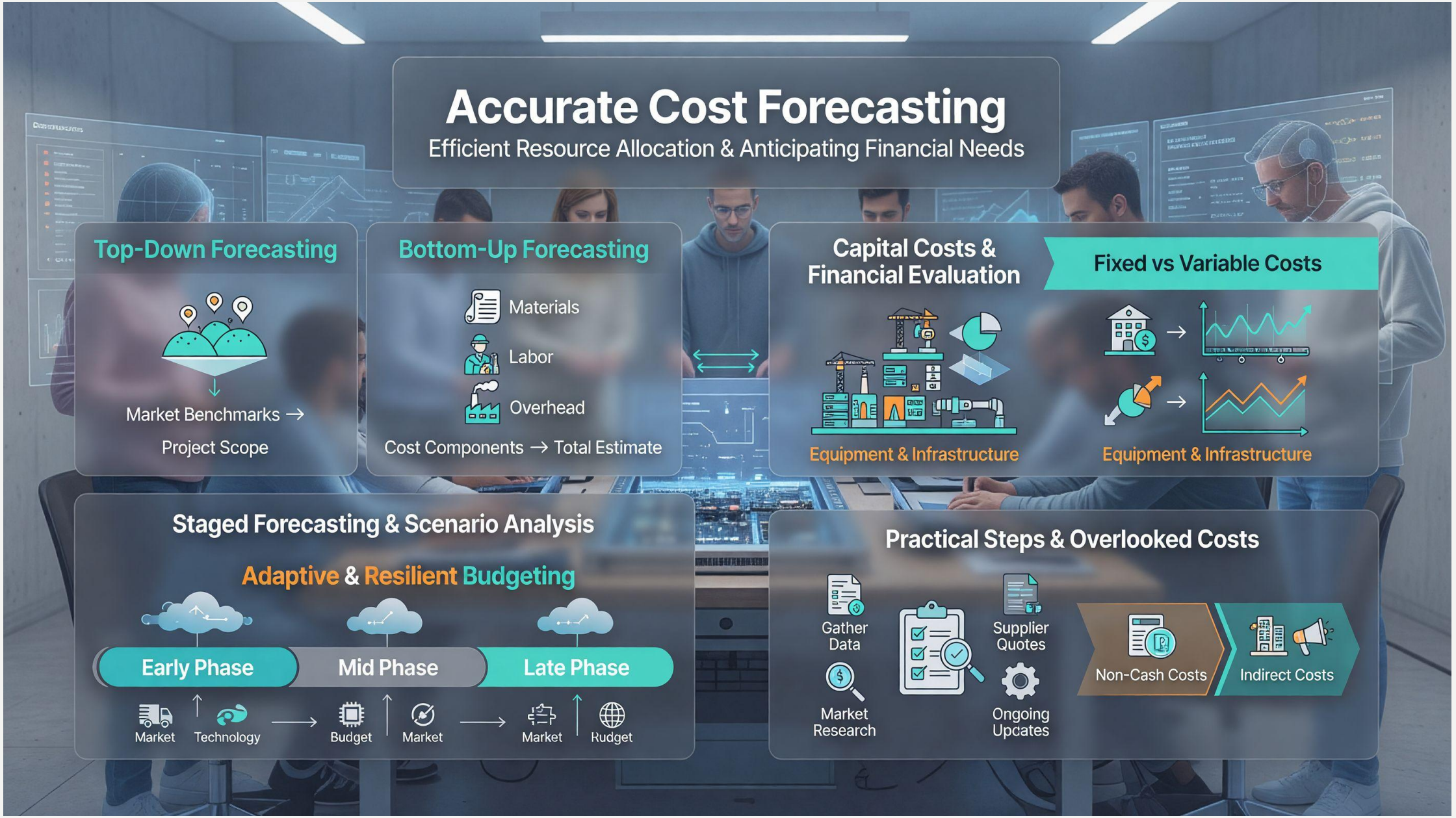
Forecasting Costs and Revenue Streams

Ir Prof Alan Lam, JP



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Methods for Forecasting Technology Project Costs



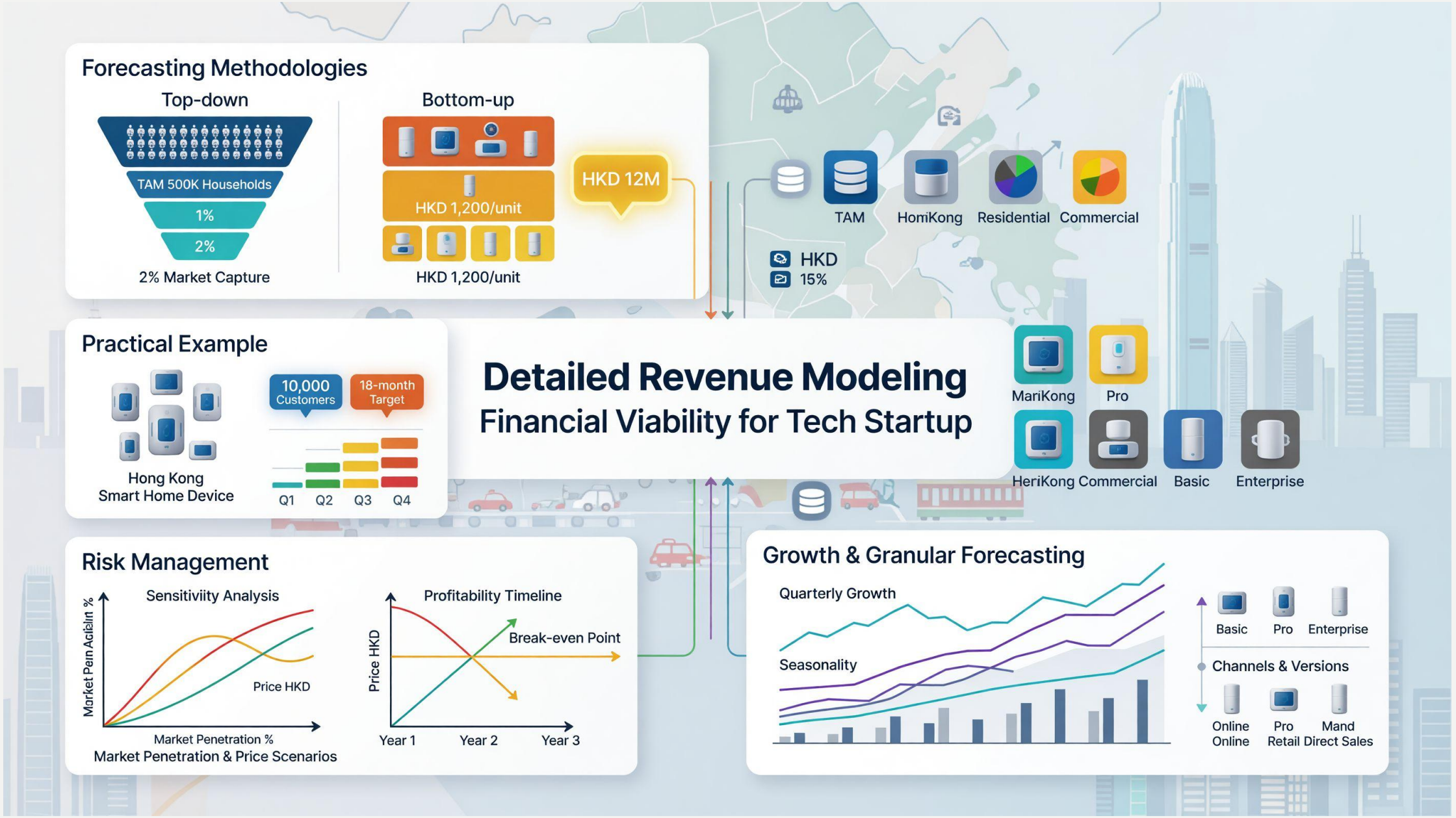
Analyzing Fixed and Variable Costs



Revenue Stream Identification



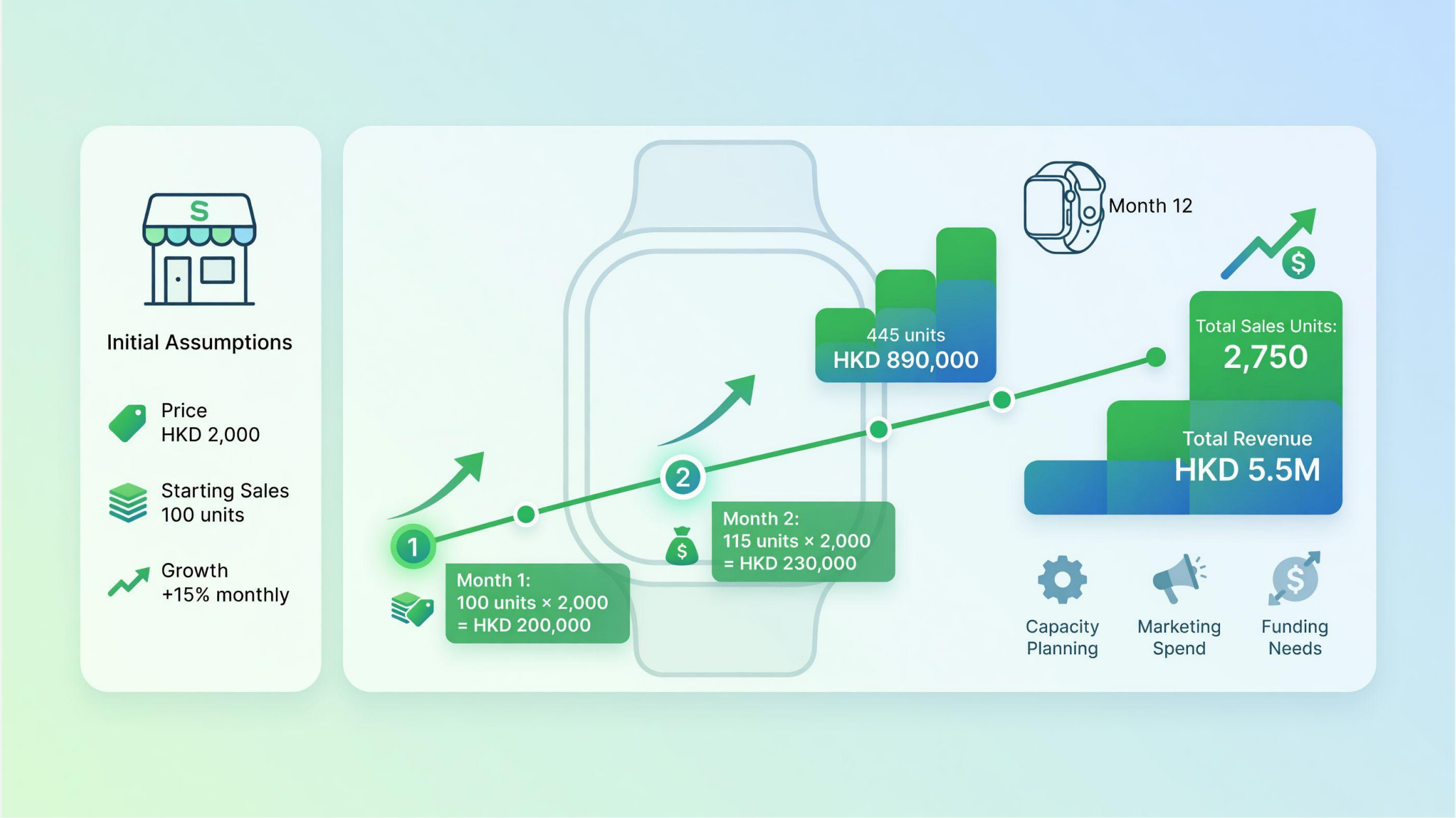
Modeling Revenue Streams from Product Sales



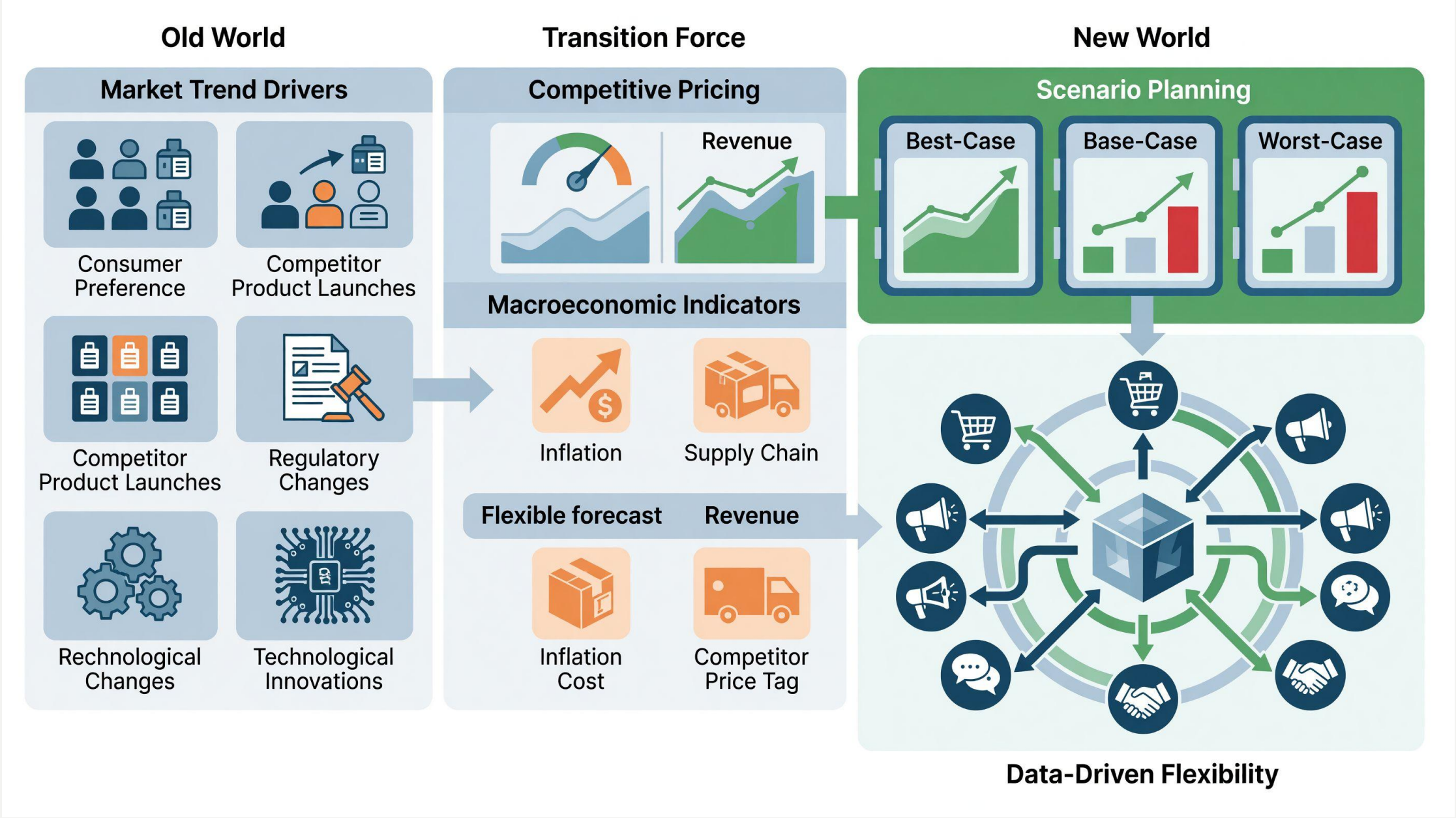
Incorporating Subscription and Licensing Revenues



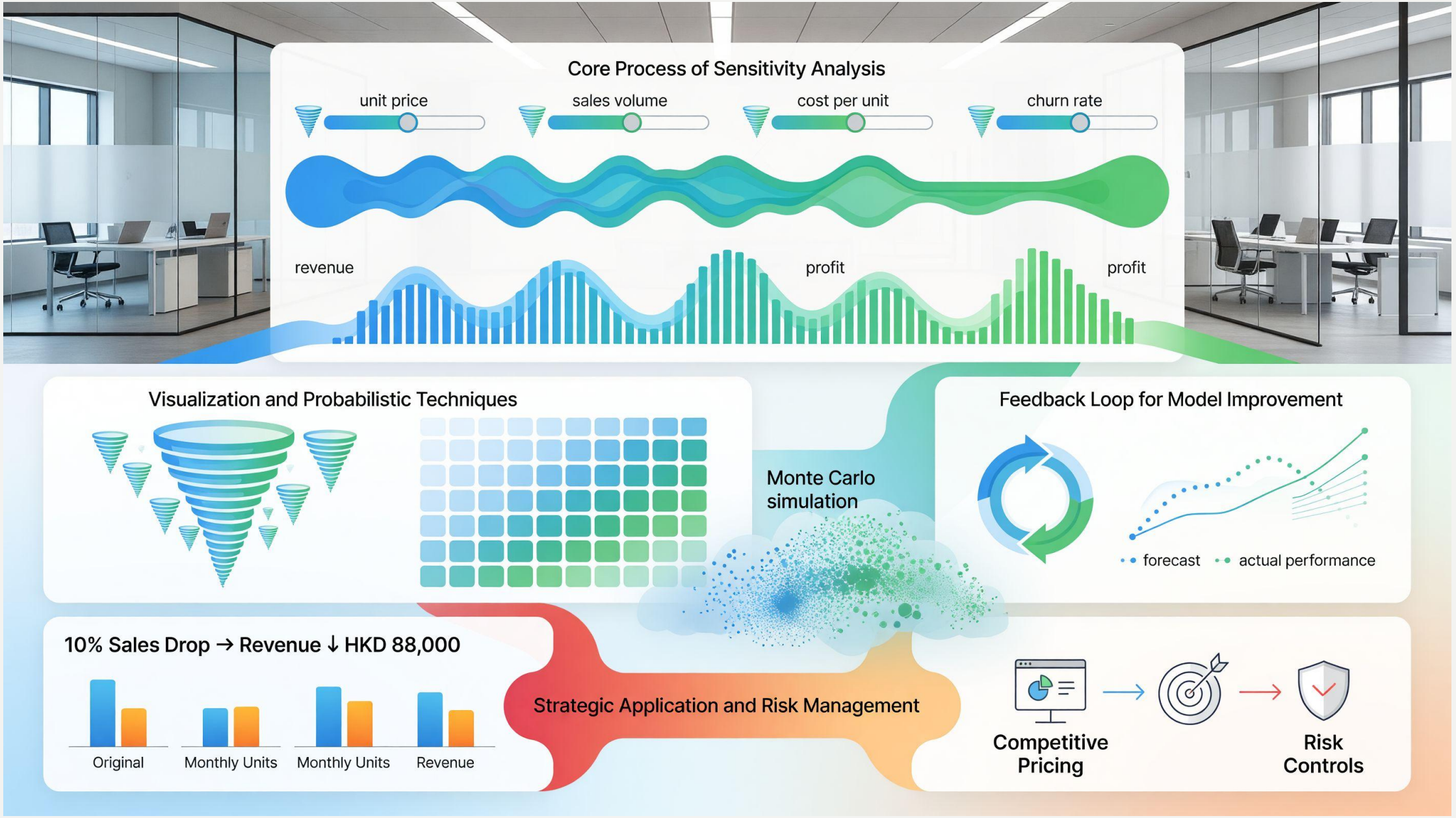
Example Problem: Forecasting Product Sales Revenue



Adjusting Forecasts According to Market Trends



Sensitivity Analysis of Forecasting Accuracy



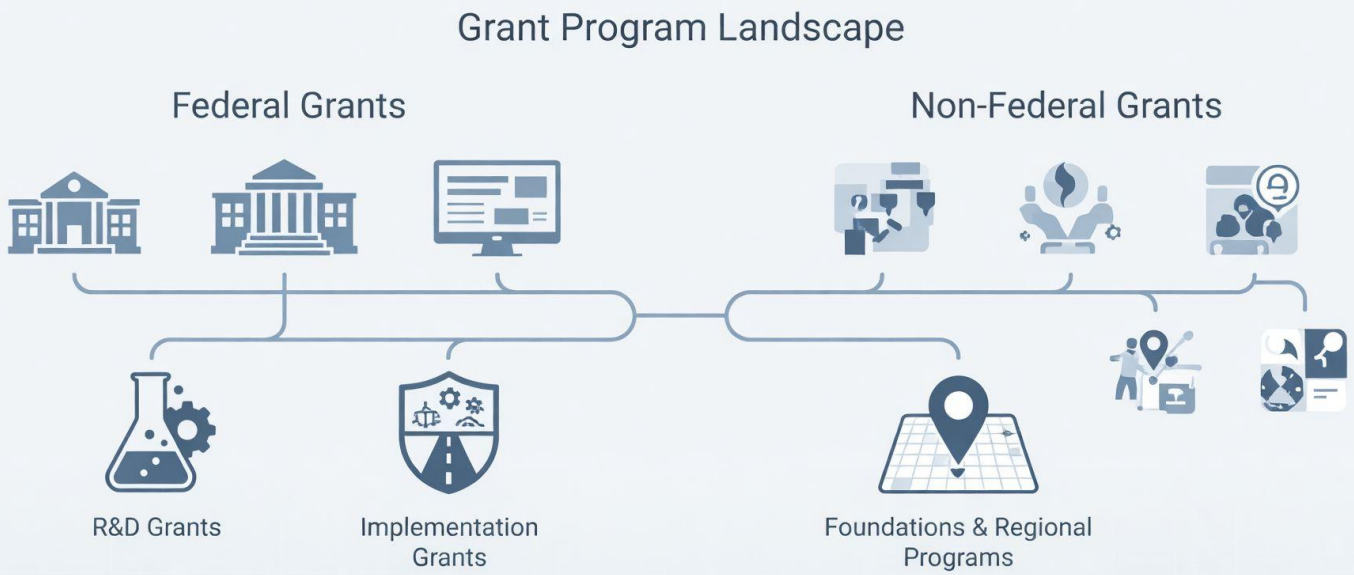
Grant Application and Utilizing Public Funding Sources

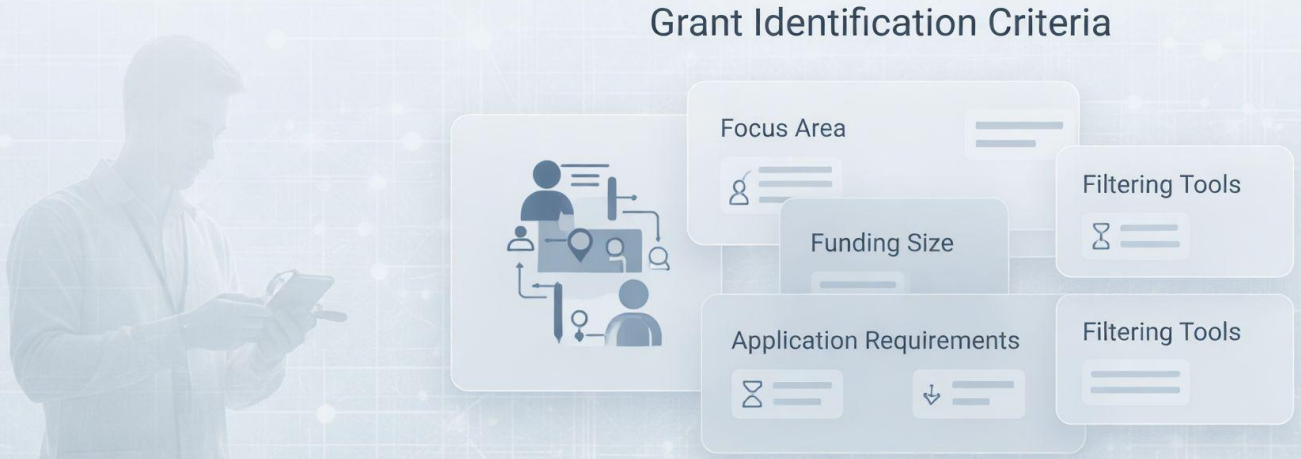
Ir Prof Alan Lam, JP



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Identifying Suitable Grant Programs





Funding Scheme in HK:



創業資本企劃: 政府資助計劃

促進貿易融資: 更多關於信保局的服務

創業預算分析: 創業預算分析表

年輕創業人士: 香港青年創業計劃

Y.E.S.向青年企業家提供的支援:

Y.E.S.提供的自僱支援服務

生意企劃:

創業資訊服務

「問問專家」業務諮詢服務

市場資訊: 香港經貿 香港行業概況 商會 領事館

聘請員工: 勞工處互動就業服務

Funding Scheme in HK:

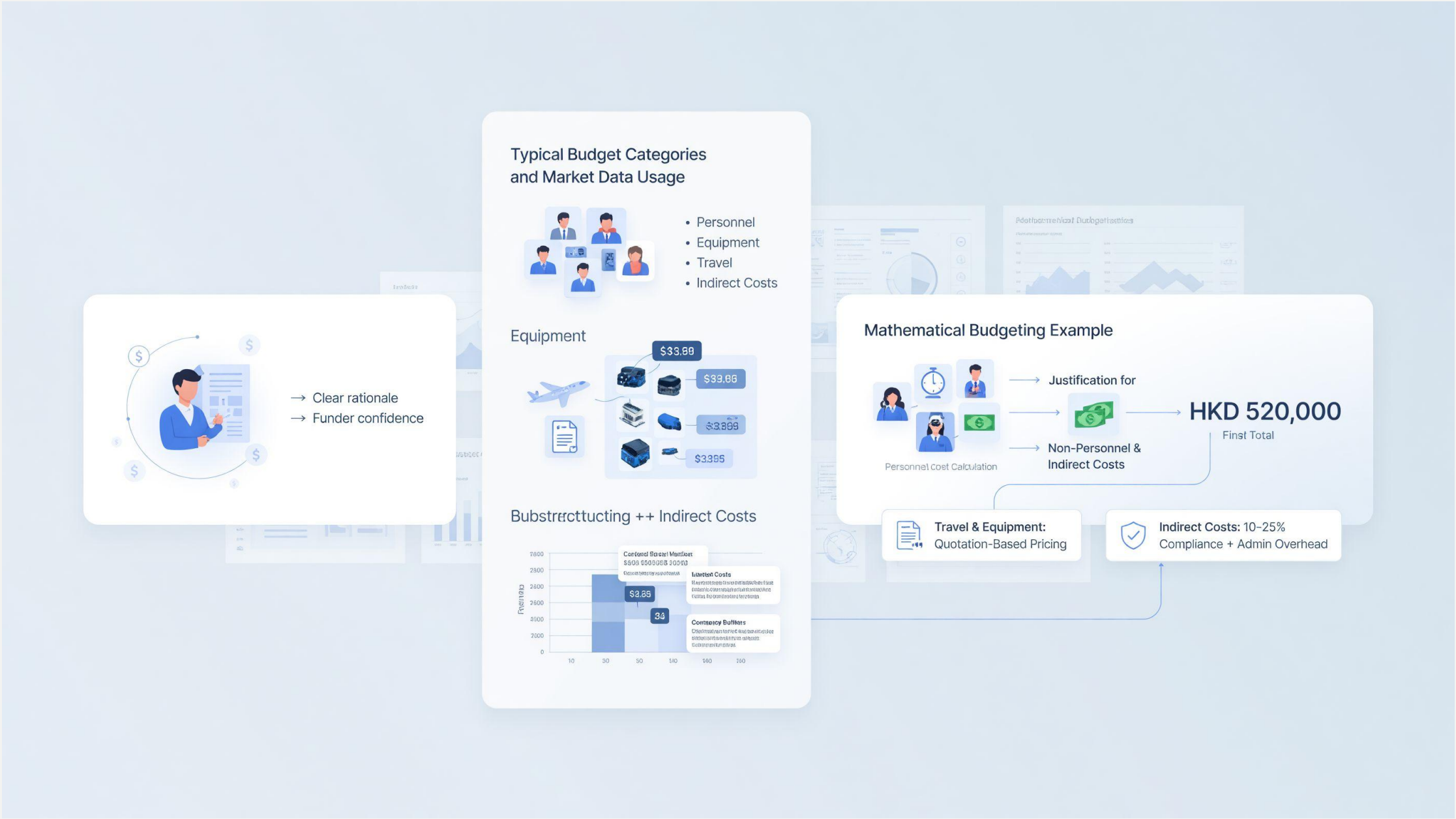




Understanding Eligibility Criteria



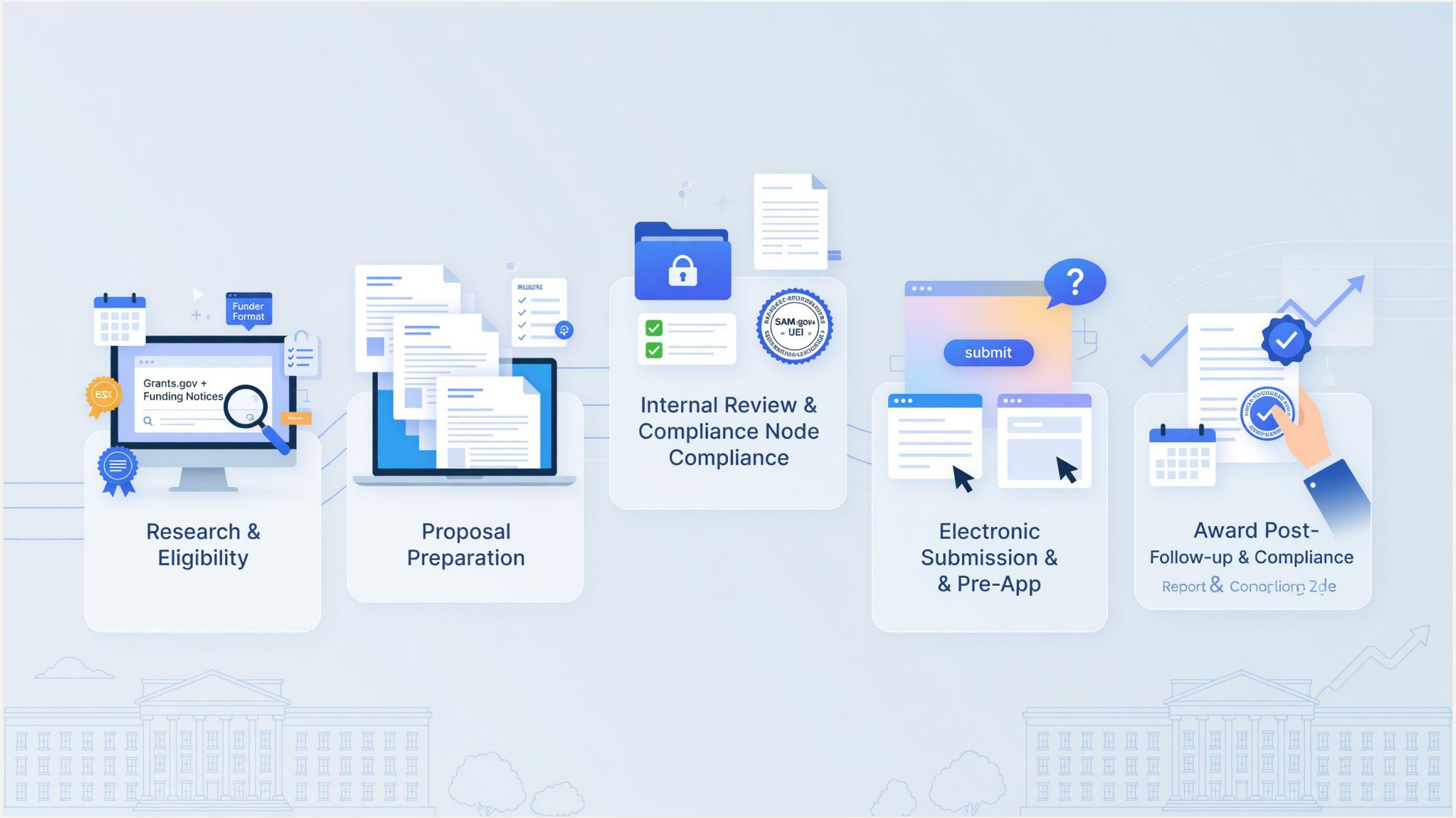
Preparing Budget Justification in Grant Proposals



Documenting Project Objectives and Outcomes



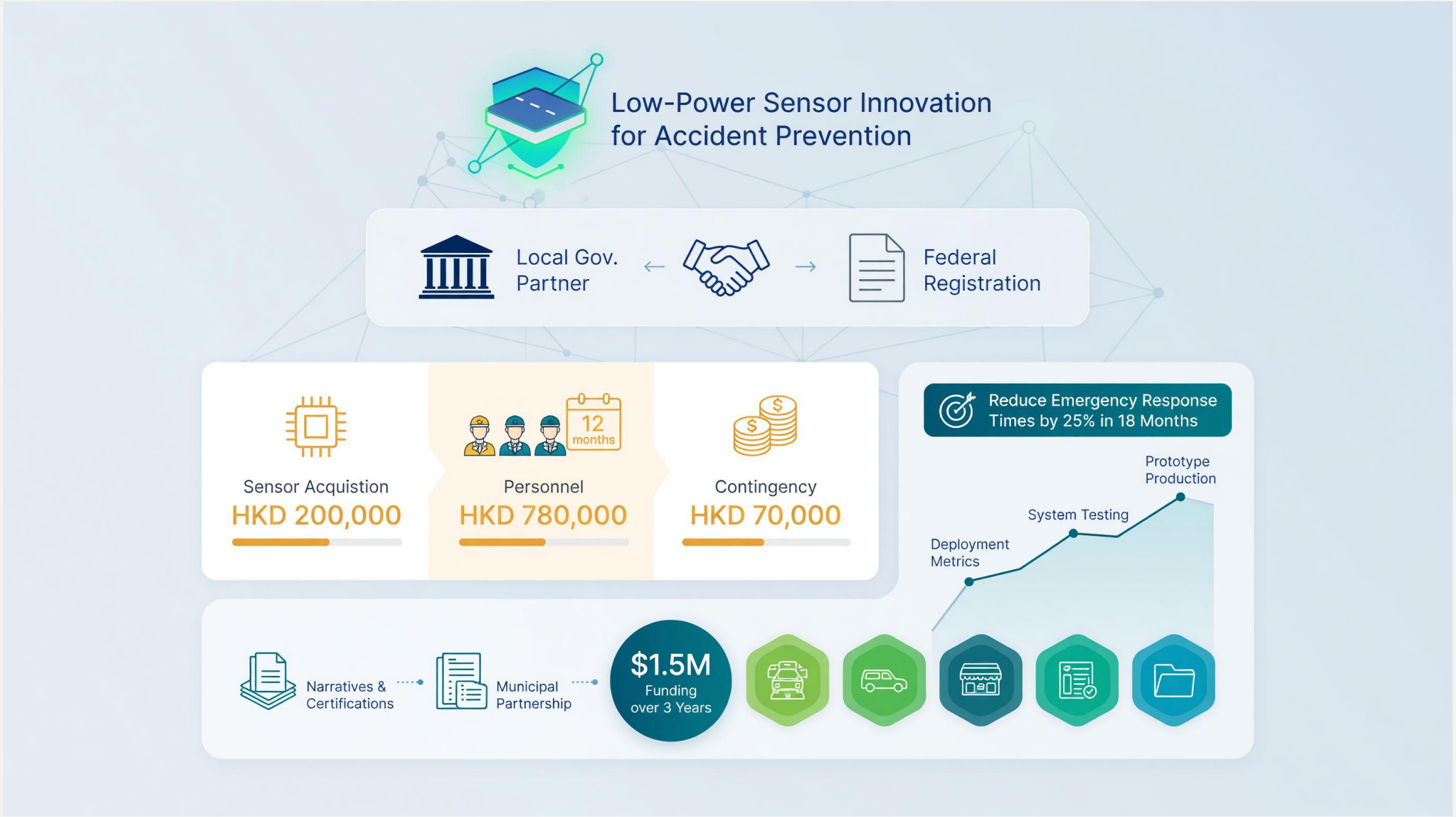
The Application Process Step-by-Step



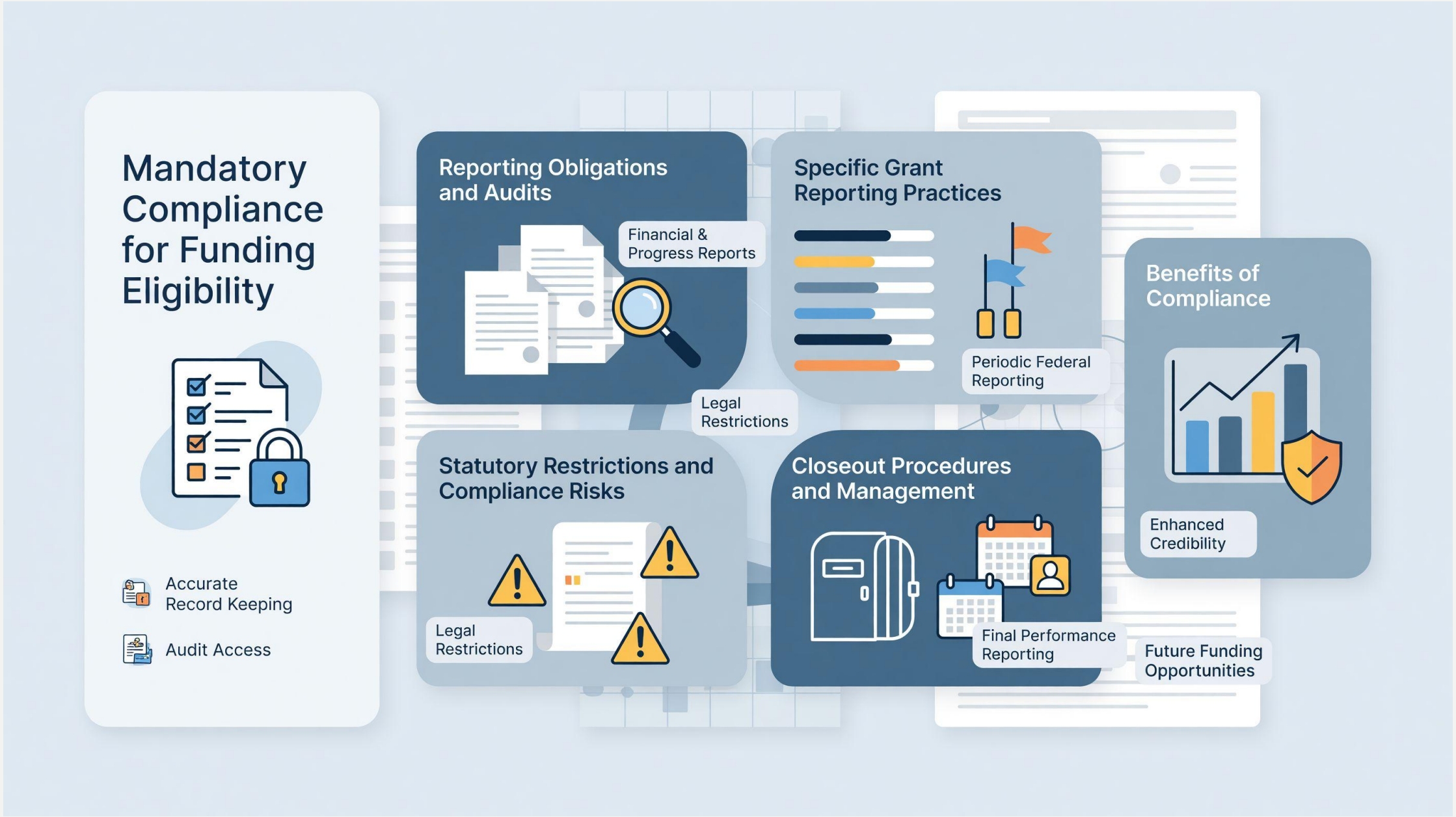
Utilizing Public Funding to Maximize Project Impact



Case Study: Successful Grant Application Example



Compliance and Reporting Requirements



Funding Your Technology-based Startup: Angel Investors, VC, Crowdfunding

Ir Prof Alan Lam, JP

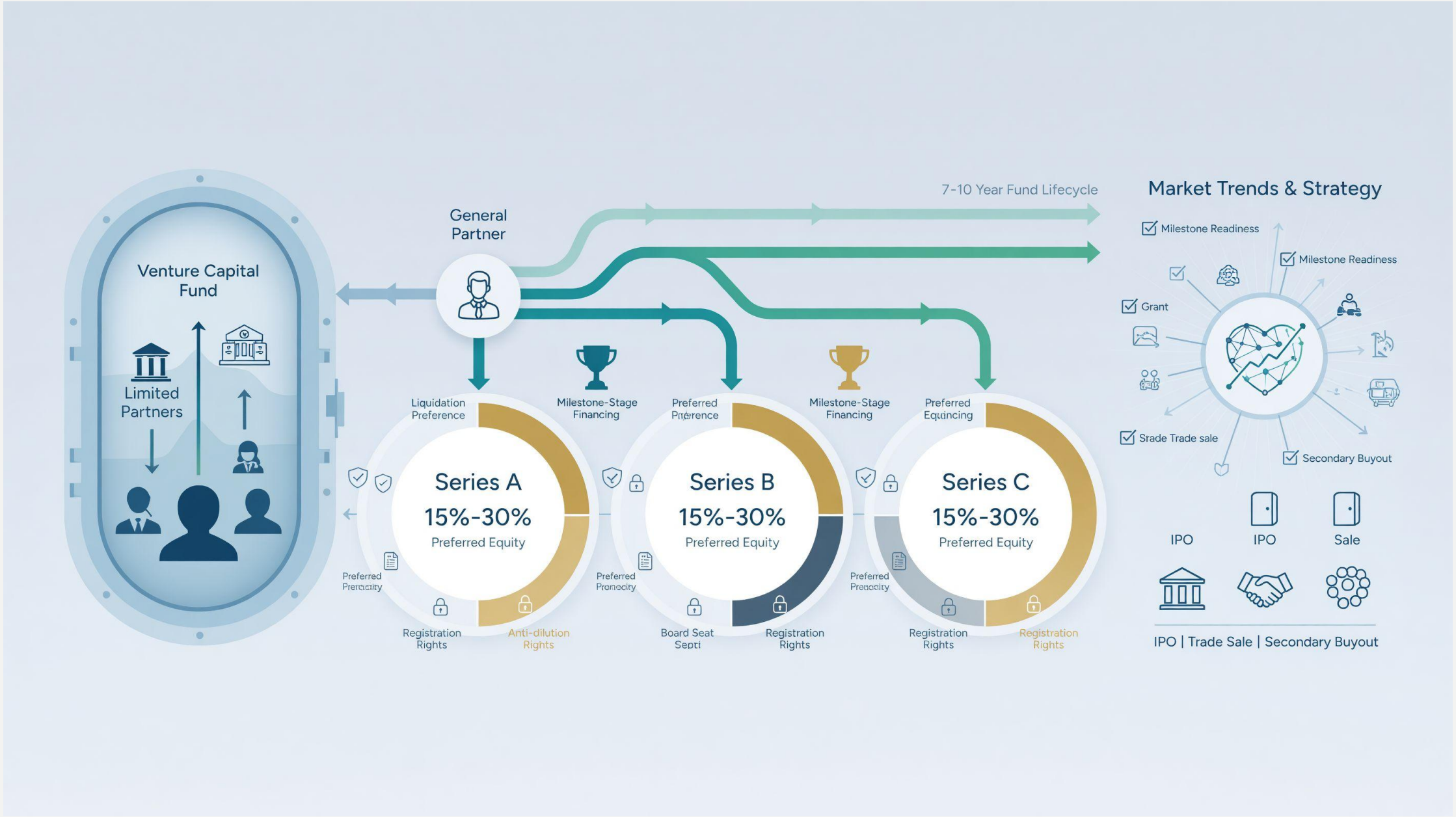


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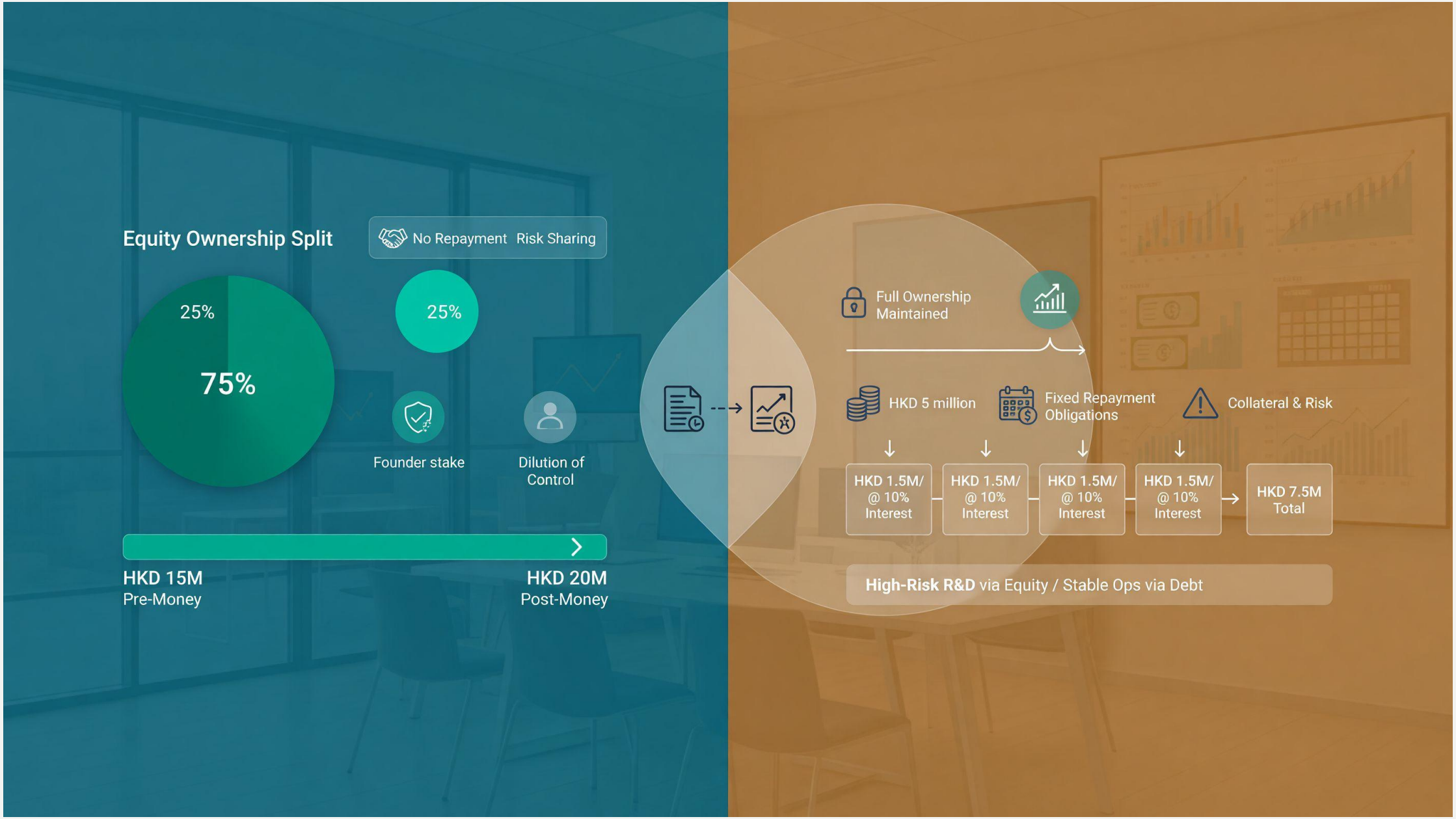
Angel Investment Fundamentals



Venture Capital Investment Structures



Equity vs. Debt Financing Options



Crowdfunding Platforms Overview



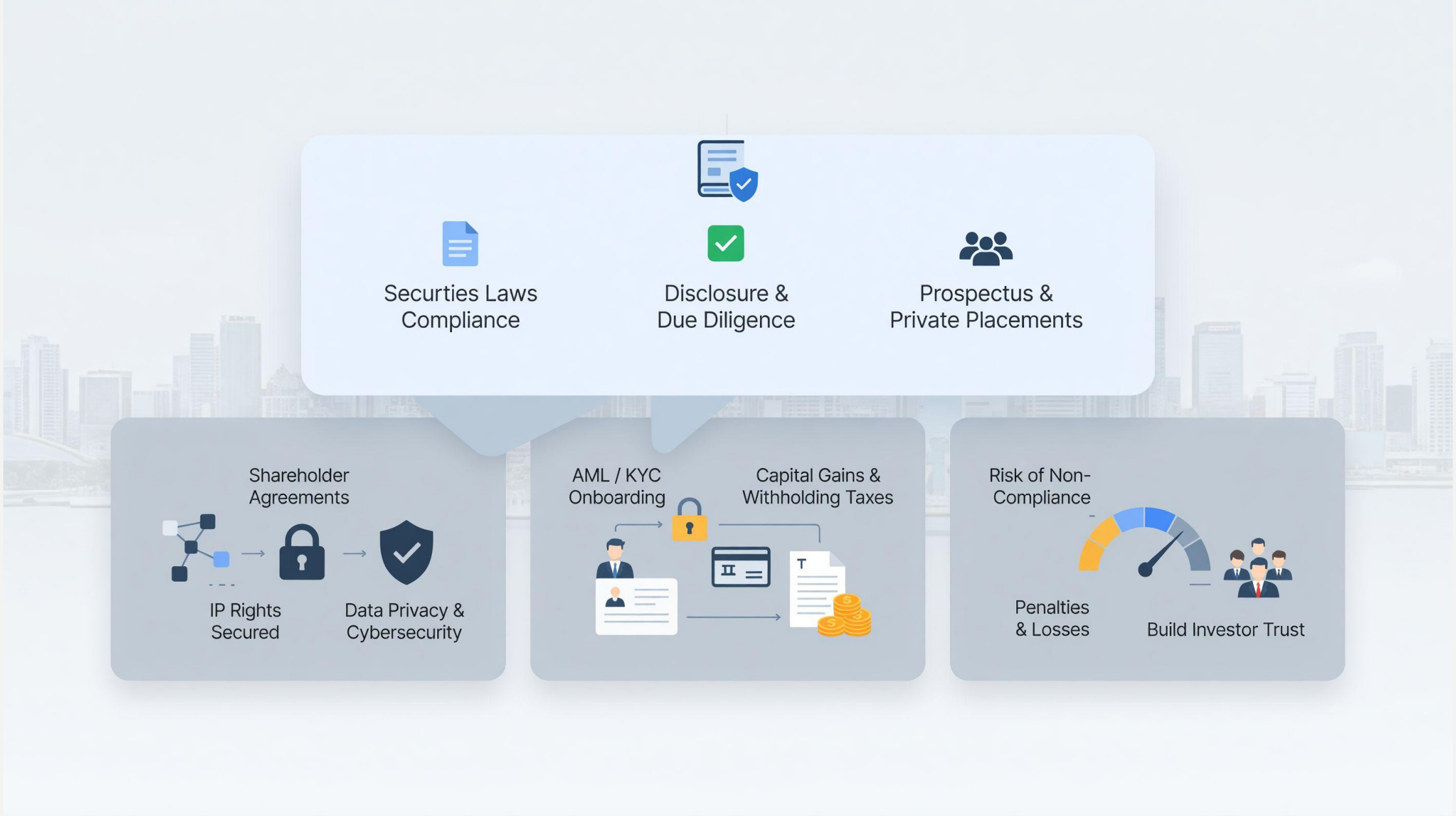
Pitching to Investors: Key Strategies



Negotiating Investment Terms and Valuation



Legal and Regulatory Considerations in Fundraising



Managing Investor Relations Post-Funding



Company Valuation, Post-Funding Financial Strategy and Growth Planning

Ir Prof Alan Lam, JP

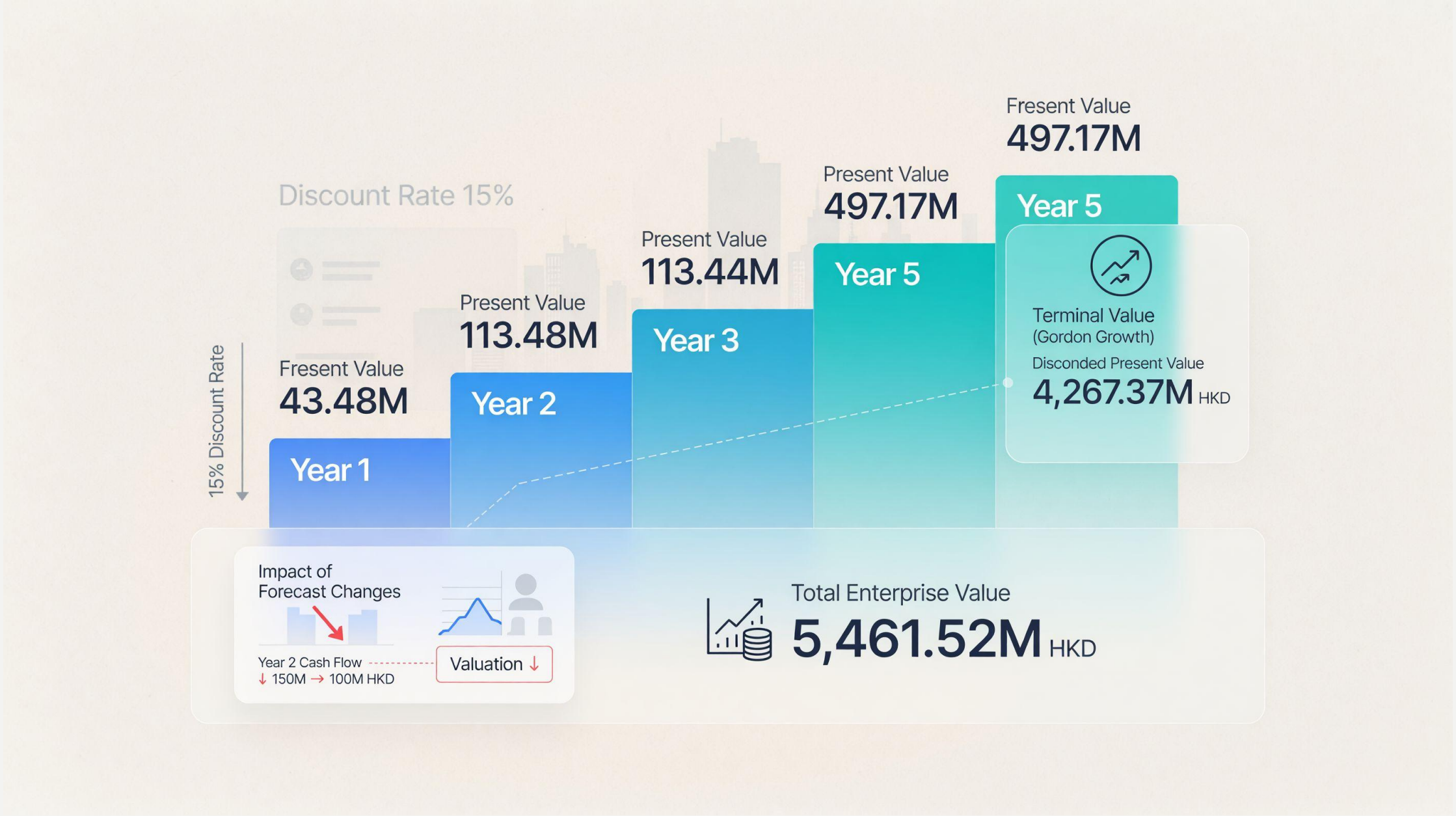


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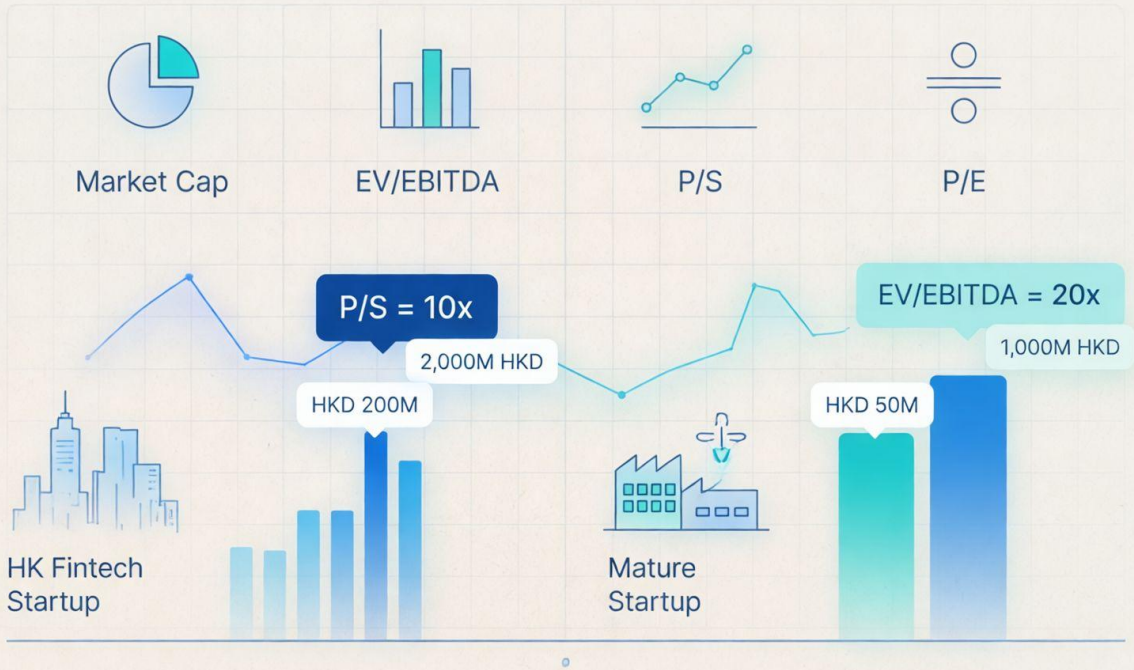
Tech Startup Valuation Models

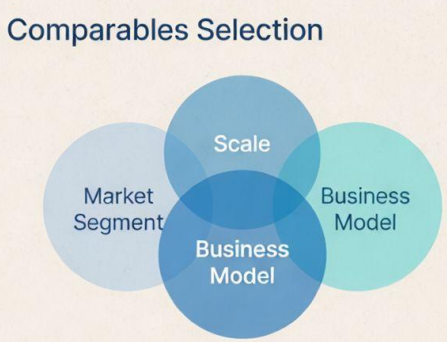


Discounted Cash Flow Analysis Explained



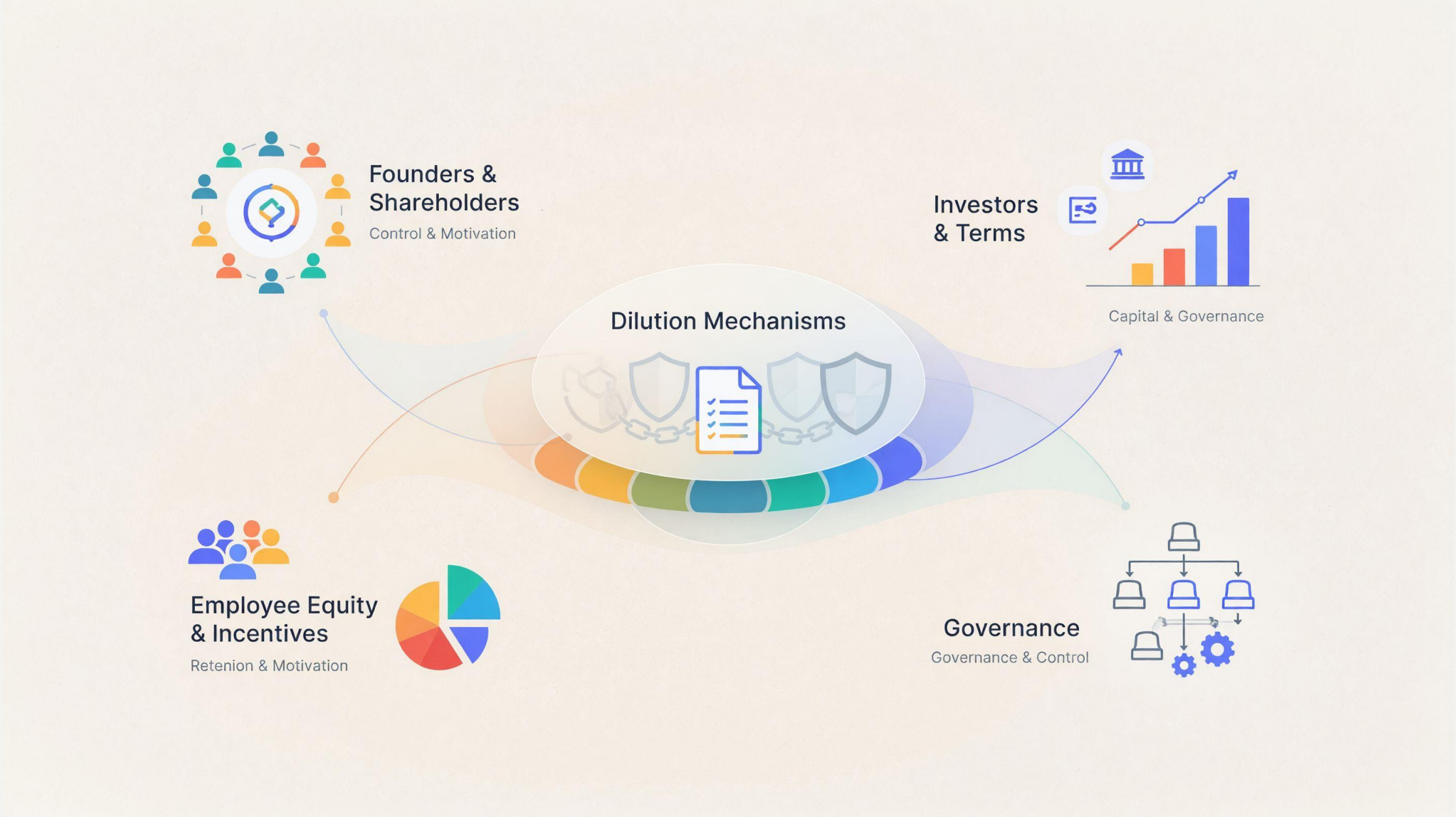
Market Comparables in Valuation



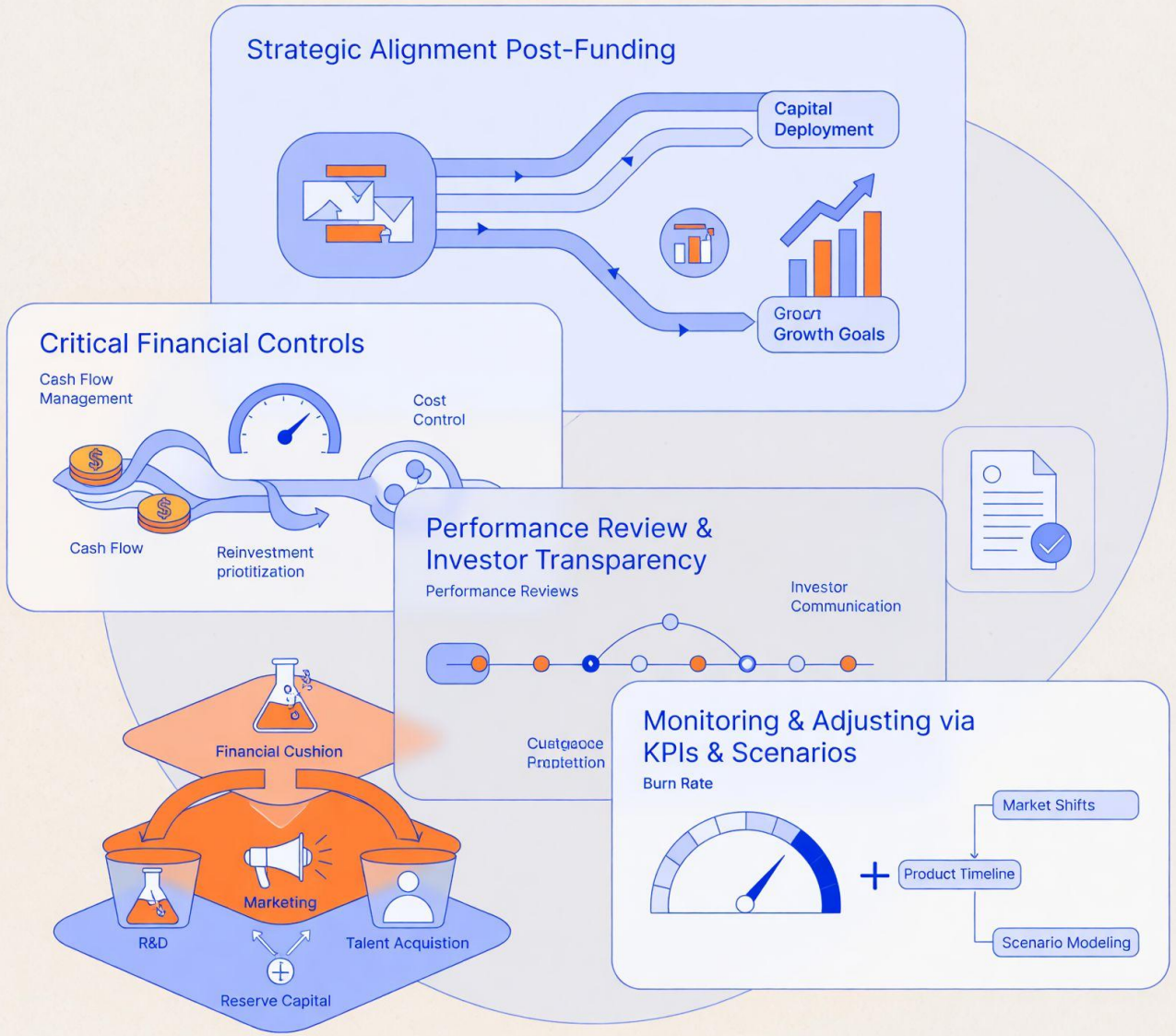




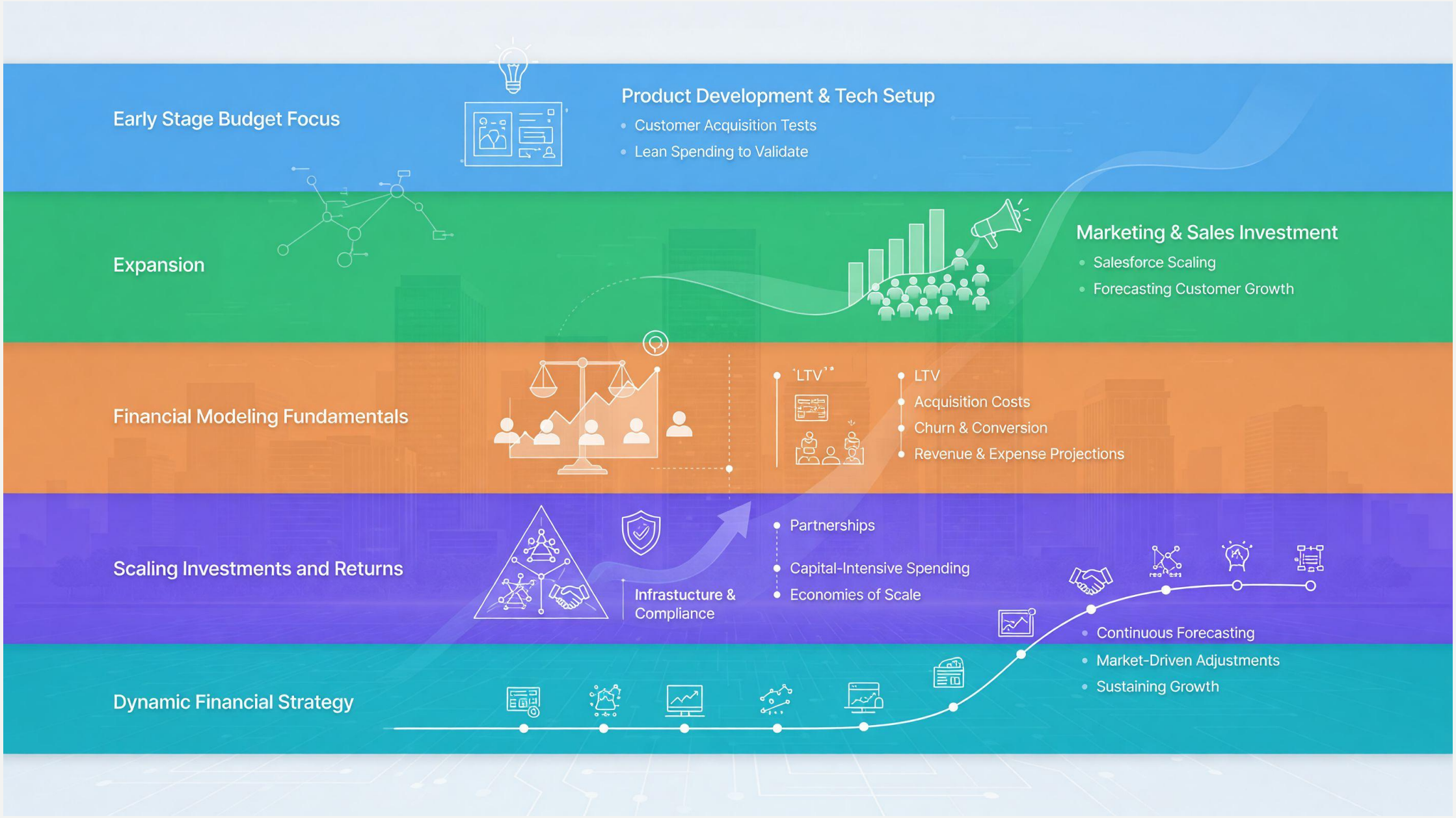
Impact of Funding on Company Equity



Financial Strategy after Funding



Budget Planning for Growth Phases



Managing Burn Rate and Runway



Strategic Planning for Scaling Operations



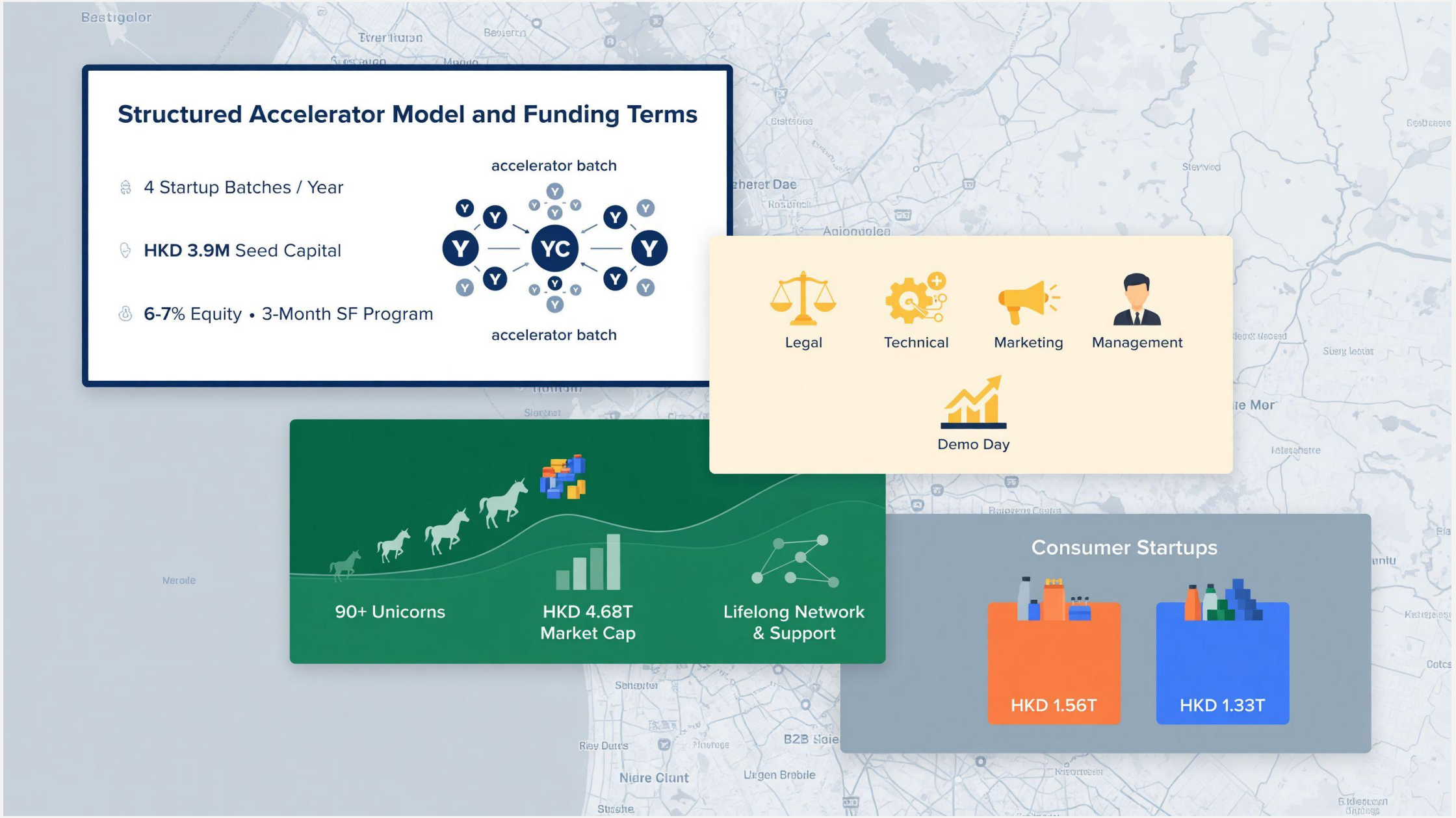
Successful Example: Y Combinator for Angel Investment and Startup Incubation

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Overview of Y Combinator's Business Model



Selection Criteria for Startups

Founding Team Strength

Complementary Skills Favored





U.S. Startups Lead

99% of Returns

Emerging High-Growth Sectors



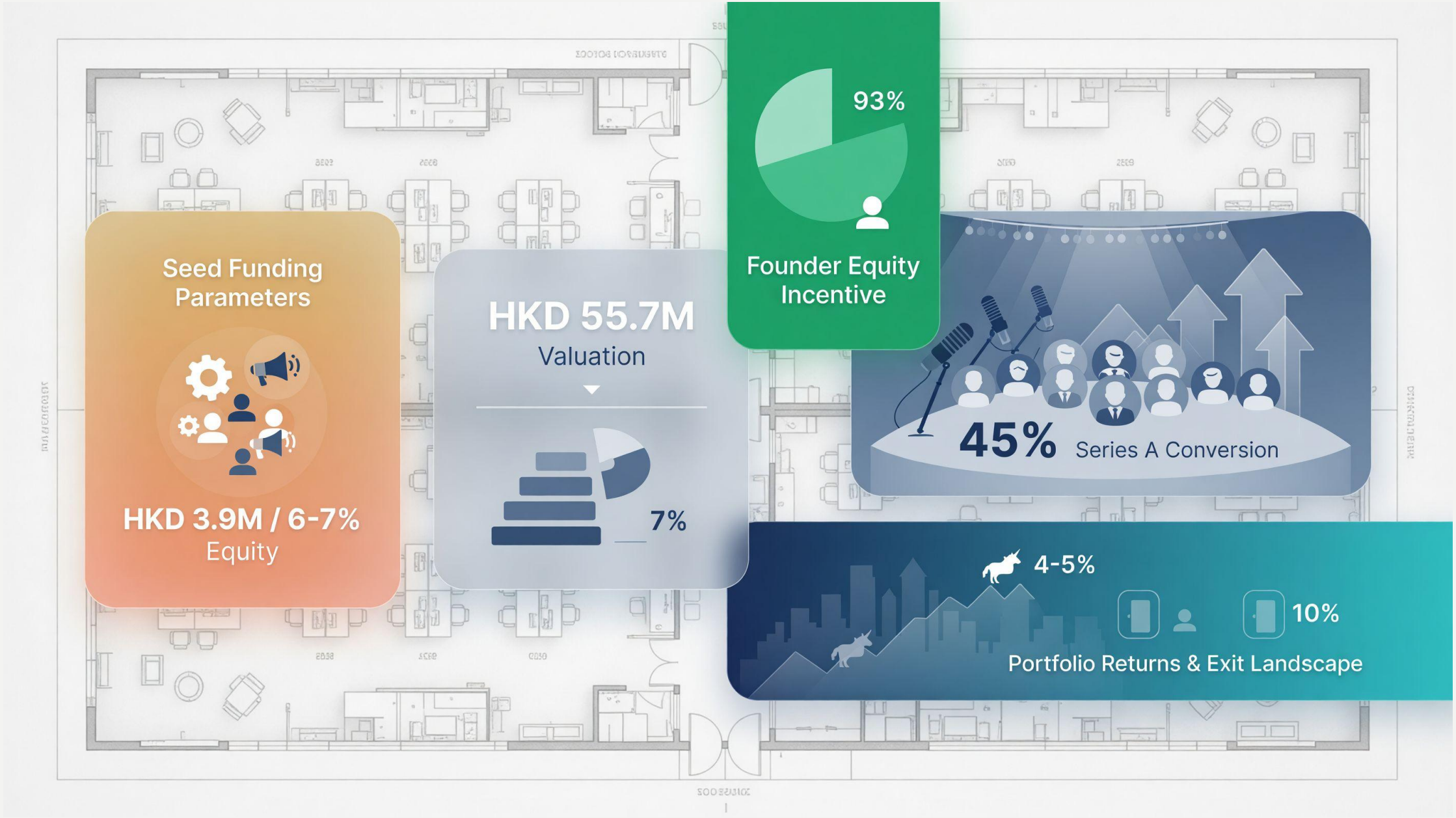
AI, Engineering, Design, Sales Tech

Scale & Funding Potential

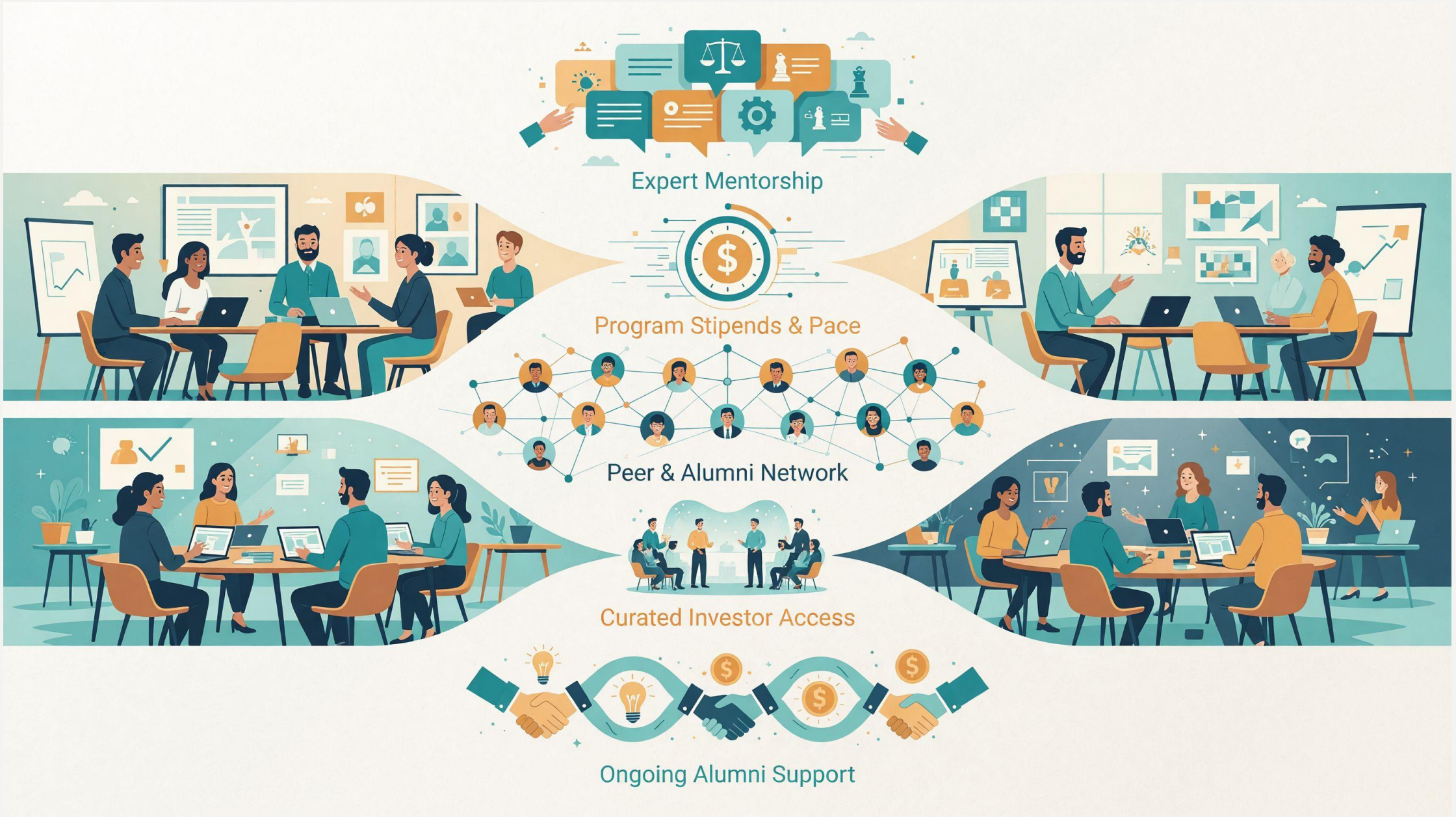
45% Reach Series A



Funding Stages and Amounts in Y Combinator



Support Services Offered to Startups



Success Metrics and Exit Strategies



Case Study: Y Combinator-backed Startup Financial Journey



Fundraising Lessons from Y Combinator



Replicating Accelerator Success in Other Contexts

Core Structural Adaptation



- Cohort Investments
- Intensive Mentorship
- Milestone Timelines

Transition Forces



Ecosystem & Community Building





Localized Selection & Sector Focus



Portfolio Returns

Diversification & High Multiples



Ongoing Operational Support



In-Class Discussion:

- Please based on the today's lecture to summarize what did you learn.
- Based on your project to work out Budget Plan and Fundraising Plan
- Make one Slide about the Business Budget and Forecast.



Startup Financing Plan:

Robotic Startup Financial Plan

Expense	Setup	Month 01	Month 02	Month 03	Month 04	Month 05	Month 06	Month 07	Month 08	Month 09	Month 10	Month 11	Month 12	Yearly	Total	Percentage
Office rental	40,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	240,000	280,000	8%
Office Setup	200,000													-	200,000	6%
Software	100,000													-	100,000	3%
Operation		20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	240,000	240,000	7%
Salary (4 headcounts)		100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000	1,200,000	1,200,000	34%
Yearly expense		50,000												50,000	50,000	1%
R&D	200,000	50,000	50,000	50,000	50,000	50,000	50,000	250,000	50,000	50,000	150,000	50,000	50,000	900,000	1,100,000	31%
Component cost											5,000	25,000	50,000	80,000	80,000	
Product cost																
Marketing													100,000	100,000	100,000	3%
Buffer budget	60,000												100,000	100,000	160,000	5%
Total Expense	600,000	240,000	190,000	190,000	190,000	190,000	190,000	390,000	190,000	190,000	295,000	215,000	440,000	2,910,000	3,510,000	
Income																
University funding	1,000,000				500,000									500,000	1,500,000	
Government funding											100,000			100,000	100,000	
Service income								50,000	50,000	50,000	50,000	50,000	50,000	300,000	300,000	
Component sales											10,000	50,000	100,000	160,000	160,000	
Product sales																
Total Income	1,000,000	-	-	-	500,000	-	-	50,000	50,000	50,000	160,000	100,000	150,000	1,060,000	2,060,000	
Profit and Loss	400,000	(240,000)	(190,000)	(190,000)	310,000	(190,000)	(190,000)	(340,000)	(140,000)	(140,000)	(135,000)	(115,000)	(290,000)	(1,850,000)		
Investment		1,000,000						1,000,000						2,000,000	2,000,000	SAFE agreement
Cashflow statement																
Cash in bank	400,000	1,160,000	970,000	780,000	1,090,000	900,000	710,000	1,370,000	1,230,000	1,090,000	955,000	840,000	550,000			

TSSSU

Income

1st $ Point by service model

1st $ Point by component sales

HKSTP

Sales Income

Investment

Budget and Income Milestones:

Sales

	Robot Sales	1st Order	2nd Order	3rd Order	4th Order			
	Order qty	100	200	500	1,000			
Fixed Cost	Setup cost	Qty	Unit Cost	Total				
	Tooling	1	300,000	300,000				
	Testing jig	10	10,000	100,000				
					400,000			
Unit Cost	Hardware	Qty	Unit Cost	Total	1st Order	2nd Order	3rd Order	4th Order
	Edge computer	1	3,000	3,000	300,000	600,000	1,500,000	3,000,000
	Components	5	100	500	50,000	100,000	250,000	500,000
	PCB	1	50	50	5,000	10,000	25,000	50,000
	SMT Assembly	1	30	30	3,000	6,000	15,000	30,000
	Production	1	200	200	20,000	40,000	100,000	200,000
	Packaging	1	20	20	2,000	4,000	10,000	20,000
	Logistics/shipment				100,000	150,000	300,000	600,000
	Sub-total				480,000	910,000	2,200,000	4,400,000
	Total with 10% Buffer				528,000	1,001,000	2,420,000	4,840,000
	Yearly fee							
Software Cost	Software	Qty	Unit Cost	Total				
	ROS	1	2,000	2,000	2,000	2,000	2,000	2,000
	App	1	5,000	5,000	5,000	5,000	5,000	5,000
Operation Cost	System	Qty	Unit Cost	Total				
	AWS cloud	1	5,000	5,000	5,000	5,000	5,000	5,000
	Total Budget				940,000	1,413,000	2,832,000	5,252,000
Income	Pricing							
	Margin	20		Revenue	4,700,000	7,065,000	14,160,000	26,260,000
	Gross Profit				3,760,000	5,652,000	11,328,000	21,008,000

3-5 Budget and Forecasting:

Yearly Forecast

	Startup	Year 01	Year 02	Year 03	Year 04	Year 05
Expense	(600,000)	(2,910,000)	(3,492,000)	(4,190,400)	(5,028,480)	(6,034,176)
Sales cost			(940,000)	(1,413,000)	(2,832,000)	(5,252,000)
Income	1,000,000	1,060,000	2,120,000	4,240,000	8,480,000	16,960,000
Sales income			4,700,000	7,065,000	14,160,000	26,260,000
P&L	400,000	(1,850,000)	2,388,000	5,701,600	14,779,520	31,933,824
Investment	-	2,000,000	2,000,000			
Dividend				8,000,000	14000000	
Cash in bank	400,000	550,000	4,938,000	2,639,600	3,419,120	35,352,944

Income

Sales

1st $ Point by product sales model

Breakeven point

Investment

Dividend

Return on investment point

Cashflow

Fundraising Plan and Strategy:

Project Fundraising							
	Valuation	Share	year 1	year 2	year 3	year 4	year 5
Angel Fund → Angel fund (SAFE with 20% discount rate)	16,000,000	12.50	2,000,000				
Series A → Series A	20,000,000	10	2,000,000				

Startup Budget and Fundraising

Reference:

- https://www.wrike.com/blog/12-steps-to-a-startup-business-plan-infographic/
- https://hbr.org/1985/05/how-to-write-a-winning-business-plan
- https://monday.com/blog/productivity/startup-budget-template/
- https://www.brex.com/journal/startup-budget
- https://www.indeed.com/career-advice/career-development/budget-for-startup



Ir Prof Alan Lam, JP
Adjunct Professor

End of Presentation
Thank You